Отчёт по практической работе №4

1. В зоне iBGP необходимо назначить на интерфейсы маршрутизаторов IP-адреса из сетей на /24.

Интерфейсы на роутере C-iBGP-102:

```
interface Ethernet0/0
 description "DMikheeva-Main-Router"
 ip address 16.1.1.2 255.255.255.252
!
interface Ethernet0/1
 description "DMikheeva-Mikrotik"
 ip address 15.1.1.1 255.255.255.252
!
interface Ethernet0/2
 description "DMikheeva-C-iBGP-15"
 ip address 13.231.1.102 255.255.255.0
!
interface Ethernet0/3
 description "DMikheeva-M-iBGP-101"
 ip address 13.231.2.102 255.255.255.0
!
interface Ethernet1/0
 description "DMikheeva-C-iBGP-28"
 ip address 13.231.3.102 255.255.255.0
!
interface Ethernet1/1
 no ip address
 shutdown
!
interface Ethernet1/2
 no ip address
 shutdown
!
interface Ethernet1/3
 no ip address
 shutdown
```

Интерфейсы на роутере C-iBGP-15:

```
interface Ethernet0/0
 description "DMikheeva-C-iBGP-102"
 ip address 13.231.1.15 255.255.255.0
!
interface Ethernet0/1
 description "DMikheeva-M-iBGP-85"
 ip address 13.164.12.15 255.255.255.0
!
interface Ethernet0/2
 description "DMikheeva-C-iBGP-23"
 ip address 13.231.4.15 255.255.255.0
!
interface Ethernet0/3
 description "DMikheeva-M-iBGP-101"
 ip address 13.231.5.15 255.255.255.0
!
interface Ethernet1/0
 no ip address
 shutdown
!
interface Ethernet1/1
 no ip address
 shutdown
!
interface Ethernet1/2
 no ip address
 shutdown
!
interface Ethernet1/3
 no ip address
 shutdown
```

Интерфейсы на роутере C-iBGP-23:

```
interface Ethernet0/0
 description "DMikheeva-C-iBGP-15"
 ip address 13.231.4.23 255.255.255.0
!
interface Ethernet0/1
 description "DMikheeva-C-iBGP-33"
 ip address 13.231.10.23 255.255.255.0
!
interface Ethernet0/2
 description "DMikheeva-M-iBGP-101"
 ip address 13.231.8.23 255.255.255.0
!
interface Ethernet0/3
 description "DMikheeva-M-iBGP-62"
 ip address 13.164.13.23 255.255.255.0
!
interface Ethernet1/0
 description "DMikheeva-M-iBGP-85"
 ip address 13.164.17.23 255.255.255.0
!
interface Ethernet1/1
 description "DMikheeva-M-iBGP-42"
 ip address 13.164.16.23 255.255.255.0
!
interface Ethernet1/2
 no ip address
 shutdown
!
interface Ethernet1/3
 no ip address
 shutdown
```

Интерфейсы на роутере C-iBGP-33:

```
interface Ethernet0/0
 description "DMikheeva-C-iBGP-28"
 ip address 13.231.7.33 255.255.255.0
!
interface Ethernet0/1
 description "DMikheeva-C-iBGP-23"
 ip address 13.231.10.33 255.255.255.0
!
interface Ethernet0/2
 description "DMikheeva-M-iBGP-101"
 ip address 13.231.9.33 255.255.255.0
!
interface Ethernet0/3
 description "DMikheeva-M-iBGP-42"
 ip address 13.164.15.33 255.255.255.0
!
interface Ethernet1/0
 no ip address
 shutdown
!
interface Ethernet1/1
 no ip address
 shutdown
!
interface Ethernet1/2
 no ip address
 shutdown
!
interface Ethernet1/3
 no ip address
 shutdown
```

Интерфейсы на роутере C-iBGP-28:

```
interface Ethernet0/0
 no ip address
 shutdown
!
interface Ethernet0/1
 description "DMikheeva-C-iBGP-102"
 ip address 13.231.3.28 255.255.255.0
!
interface Ethernet0/2
 description "DMikheeva-C-iBGP-33"
 ip address 13.231.7.28 255.255.255.0
!
interface Ethernet0/3
 description "DMikheeva-M-iBGP-101"
 ip address 13.231.6.28 255.255.255.0
!
interface Ethernet1/0
 no ip address
 shutdown
!
interface Ethernet1/1
 no ip address
 shutdown
!
interface Ethernet1/2
 no ip address
 shutdown
!
interface Ethernet1/3
 no ip address
 shutdown
!
```

Интерфейсы на роутере M-iBGP-101:

```
[admin@DMikheeva-M-iBGP-101] > ip address pr
Flags: X - disabled, I - invalid, D - dynamic
 #   ADDRESS            NETWORK         INTERFACE
 0   ;;; "DMikheeva-C-iBGP-102"
     13.231.2.101/24    13.231.2.0      ether1
 1   ;;; "DMikheeva-C-iBGP-28"
     13.231.6.101/24    13.231.6.0      ether2
 2   ;;; "DMikheeva-C-iBGP-15"
     13.231.5.101/24    13.231.5.0      ether3
 3   ;;; "DMikheeva-C-iBGP-23"
     13.231.8.101/24    13.231.8.0      ether4
 4   ;;; "DMikheeva-C-iBGP-33"
     13.231.9.101/24    13.231.9.0      ether5
```

2. Настроить Internal BGP в зоне iBGP.

Трассировка от роутера C-iBGP-102 до роутера C-iBGP-23:

```
DMikheeva-C-iBGP-102#traceroute 13.231.4.23
Type escape sequence to abort.
Tracing the route to 13.231.4.23
VRF info: (vrf in name/id, vrf out name/id)
  1 13.231.2.101 4 msec 2 msec 4 msec
  2 13.231.5.15 4 msec 3 msec 2 msec
  3 13.231.4.23 3 msec 7 msec *
```

Трассировка от роутера C-iBGP-33 до роутера C-iBGP-102:

```
DMikheeva-C-iBGP-33#trace 13.231.2.102
Type escape sequence to abort.
Tracing the route to 13.231.2.102
VRF info: (vrf in name/id, vrf out name/id)
  1 13.231.9.101 3 msec 3 msec 2 msec
  2 13.231.2.102 2 msec 5 msec *
```

3. Настроить External BGP между роутерами C-iBGP-102, Router, M-OSPF-10 и C-RIP-1.

Трассировка от роутера C-iBGP-15 до роутера C-RIP-10:

```
DMikheeva-C-iBGP-15#trace 186.208.48.2
Type escape sequence to abort.
Tracing the route to 186.208.48.2
VRF info: (vrf in name/id, vrf out name/id)
  1 13.231.1.102 2 msec 2 msec 1 msec
  2 15.1.1.2 17 msec 12 msec 7 msec
  3 18.1.1.1 [AS 300] 3 msec 6 msec 3 msec
  4 140.116.9.2 [AS 200] 11 msec 6 msec 7 msec
  5 93.160.249.2 [AS 200] 3 msec 9 msec 7 msec
  6 93.160.240.2 [AS 200] 23 msec 8 msec 22 msec
  7 186.208.48.2 [AS 200] 10 msec 13 msec *
```

Трассировка от роутера C-iBGP-28 до роутера C-OSPF-5:

```
DMikheeva-C-iBGP-28#trace 12.208.16.2
Type escape sequence to abort.
Tracing the route to 12.208.16.2
VRF info: (vrf in name/id, vrf out name/id)
  1 13.231.3.102 1 msec 4 msec 1 msec
  2 15.1.1.2 2 msec 4 msec 7 msec
  3 12.229.10.2 [AS 300] 6 msec 3 msec 15 msec
  4 12.229.10.6 [AS 300] 10 msec 17 msec 27 msec
  5 12.208.14.2 [AS 300] 14 msec 6 msec 5 msec
  6 12.208.16.2 [AS 300] 25 msec 11 msec *
```

Трассировка от роутера M-iBGP-101 до роутера C-R-2-3:

```
[admin@DMikheeva-M-iBGP-101] > tool trace 182.92.3.2
 # ADDRESS                          LOSS SENT    LAST     AVG    BEST   WORST
 1 13.231.5.15                        0%    7   1.9ms     1.8     1.2     2.9
 2 13.231.1.102                       0%    7   2.9ms     1.9     1.5     2.9
 3 16.1.1.1                           0%    7   1.9ms     2.6     1.4       6
 4 10.101.0.1                         0%    7   9.2ms     3.2     1.8     9.2
 5 182.92.3.2                         0%    7   2.8ms     6.7     2.8    16.7
```

Трассировка от роутера C-iBGP-102 до компьютера PC19: